



In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course Number : PHA F419
Course Title : Herbal Drug Technology
Instructor-in-Charge : Prof. Paul Atish Tulshiram
Instructor(s) : NA
Tutorial/Practical Instructors : NA

- 1. Course Description:** Operations in the herbal drug industry, the quality of raw material, guidelines for quality of herbal drugs, classical ayurvedic and conventional formulations, nutraceuticals, herbal cosmetics, herbal excipients, etc. Good Manufacturing Practices (GMP), patenting, and regulatory issues of herbal drugs.
- 2. Scope and Objective:** This course will give a basic understanding of the herbal drug industry, the quality of raw material, and knowledge of the WHO and ICH guidelines for evaluation of herbal drugs, herbal cosmetics, natural sweeteners etc.

3. Course learning outcomes:

CLO1. Students will be able to work in the area of herbal drug technology with an in-depth understanding of the industry and their products.

CLO2. Students will understand Patenting & Regulatory requirements of natural products

CLO3. Students will learn about Indian Systems of Medicine

4. Text Books and references:

TB 1. HERBAL DRUG TECHNOLOGY by VM Shinde & KS Bodas, 2019

TB 2. Textbook of Pharmacognosy by Trease & Evans, 16th Edition, 2012

RB 1. Pharmacopoeial standards for Ayurvedic Formulation (Council of Research in Indian Medicine & Homeopathy)

RB 2. Mukherjee, P.W. Quality Control of Herbal Drugs: An Approach to Evaluation of Botanicals.
Business Horizons Publishers, New Delhi, India, 2002

Guidelines 1: <https://www.who.int/publications/i/item/9789241594448>



5. Course Format & Weekly Workflow

Week no.	Module no.	Description	Reference	Learning Outcome
1	I	Herbs as raw materials	TB-1	Students will learn the basics of herbal raw materials
2	II	Biodynamic Agriculture	TB-1	Students will learn the organic farming, pest management etc
3	III	Indian Systems of Medicine	TB-1& 2	Students will learn about the basic principles involved in Ayurveda, Siddha, Unani and Homeopathy
4	IV	Indian Systems of Medicine	TB-1& 2	Students will learn about the Preparation and standardization of Ayurvedic formulations
5	V	Nutraceuticals	TB-1	Students will learn about the general aspects, Market, growth, scope and types of products available in the market
6	VI	Herbal-Drug and Herb-Food Interactions	TB-1& 2	Students will learn about the possible side effects and interactions: Hypercium, kava-kava, Ginko biloba, Ginseng, Garlic, Pepper & Ephedra
7	VII	Herbal Cosmetics	TB-1	Students will learn the herbal based skin care, hair care and oral hygiene products.
8	VIII	Herbal excipients	TB-1	Students will learn the herbal excipient of natural origin
9	IX	Herbal formulations	TB-1 & RB-2	Students will learn about conventional herbal formulations like syrups, mixtures and tablets and Novel dosage forms like phytosomes
10	X	Evaluation of Drugs	Guideline 1	Students will learn about WHO & ICH guidelines for the assessment of herbal drugs & stability testing of herbal drugs.
11	XI	Patenting & Regulatory requirements of natural products	TB-1 & RB-2	Students will learn about Patenting aspects of Traditional Knowledge and Natural Products
12	XII	Regulatory Issues	TB-1 & RB-2	Students will learn about regulations in India (ASU DTAB, ASU DCC)
13	XIII	General Introduction to the Herbal Industry	TB-1	Students will learn about a brief account of plant-based industries and institutions
14	XIV	Schedule T – Good Manufacturing Practice of Indian systems of medicine	TB-1	Students will learn about the Components of GMP (Schedule – T) and its objectives



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani

Pilani Campus

6. Evaluation Scheme:

Component	Weightage (%)	Duration	Date & Time	Comments
Mid-Semester Test	30	90 min.	As per timetable	Close Book
Comprehensive Examination	40	180 min.	As per timetable	Close Book
Continuous assessment				
Quiz	10	20 min	TBA	Close Book
Assignment	10	2 weeks	TBA	Open Book
Seminar	10	20 min	TBA	Open book

Criterion for NC: *Intentional Absence for any of the above evaluation components will attract an award of NC. [30% of the average of the top three performers or 40% of median, whichever is lower, or a suitable modification at the discretion of the IC with the concurrence of the Department]*

7. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

8. Chamber Consultation Hour: To be announced in class.

9. Make-up Policy: Make-Ups are not given as a routine. It is solely dependent upon the GENUINENESS OF THE CIRCUMSTANCES under which a student fails to appear in a scheduled evaluation component. In such circumstances, prior permission should be obtained from the Instructor-in-Charge. IN NO CASE THE MAKE-UP APPLICATION BE SLIPPED INSIDE THE CHAMBER OF THE INSTRUCTOR-IN-CHARGE. However, the decision of the Instructor-in-Charge in the above matter will be final.

Instructor-in-charge
Course No. PHA F419